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Economics of poplar short rotation coppice plantations on marginal land in Germany

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ABSTRACT

Although there is a need for biomass and a potential for short rotation coppice (SRC), farmers hesitate to establish SRC, even on marginal agricultural land on which annual crops show low productivity. Probably the most important factor explaining this reluctance might be the uncertain economic prospects of the cultivation of SRC. Therefore, the aim of this study is to analyse the economy of a typical SRC supply chain by calculating the annuities which can be expected by German farmers who establish SRC on their marginal land.

The result shows that the yearly annuity of a 20-year SRC cultivation is about 70 € y⁻¹ ha⁻¹ when poplar SRC is harvested every 4 years with a forage harvester (one-step system). The result includes the establishment, cultivation and transport of the fresh wood chips to a plant 50 km away. However, this result is not competitive with the result of annual crops (226–462 € y⁻¹ ha⁻¹) and is also lower than the CAP subsidy payments that farmers receive from the EU (300 € y⁻¹ ha⁻¹). To achieve higher annuities, four options were analysed possibly leading either to higher biomass yields or to higher market prices (extension of rotation cycle, implementation of irrigation, technical drying of fresh wood chips, using a two-step harvesting system). The implementation of drip irrigation to increase biomass yield turned out to be uneconomic. An extension of the rotation cycle from 4 to 5 years can be recommended as it leads to an annuity of 255 € y⁻¹ ha⁻¹ (instead of 69 € y⁻¹ ha⁻¹). Results also show that the technical drying of chips using (cheap) surplus heat can be very profitable if the added value is reflected in higher market prices. Furthermore, it is shown that the use of an alternative two-step harvesting system with natural interim drying of the rods can be an attractive option for farmers to increase the annuity of their SRC.

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1. Introduction

Driven by concerns about global warming and striving for energy independence, the European Union has set a 20% target for the overall share of energy from renewable sources by 2020 [1]. At the same time, the aim is a transition to a low-

carbon energy economy, while an increase in future energy demand is expected [2,3].

Among renewable energy sources biomass from sustainably managed resources (especially woody biomass) plays an important role in displacing fossil fuels [4–6], due to its ability to capture carbon and store energy, and due to other

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environmental benefits such as a higher retention of nitrogen [7]. In contrast to wind and solar, biomass can provide base load capacity to the grid.

Short rotation coppices (SRC) are seen as an option to produce additional woody biomass efficiently in a short time and in a sustainable way [8,9] without competing with biomass resources from forests.

SRC are well suited for biomass production because of the rapid juvenile growth of their trees and their high biomass yields [10]. Fast growing tree species like *Populus* spp. and their hybrids (in Germany particularly “Max”, *Populus maximowiczii* × *Populus nigra*) are easy to propagate through vegetative cuttings and can be grown under a wide variety of site and climatic conditions [10,11]. They are cultivated not only on arable cropland, but also on marginal agricultural land [12,13] which is poorly suited to field crops because of low crop productivity due to climatic limitations [13] or which is unprofitable because of other reasons, e.g. due to small field sizes. Often, these sites are suitable for SRC if water availability is sufficient [14,15]. In this study, marginal agricultural lands are focused on for energy production only in order to avoid competition with land that is better suited for food production. The plantations can be harvested after 2–5 years. As the trees maintain the capacity to sprout even after several cuttings, the same plantation can be harvested several times over a 20–30 years period [16,11] before it is re-planted or returns to its former agricultural land use.

Studies have already analysed the potential of SRC in terms of biomass yields and land availability in different countries [17–20,8,3]. For the Federal Republic of Germany, Aust et al. (2013) [15] recently published a study analysing land availability and the potential biomass production of poplar and willow SRC. Taking several restrictions into account, the authors came to the result that at least 680,000 ha (ha) of marginal cropland might be suitable for SRC in Germany [15], which means there would be a large potential for the cultivation of SRC. However, only approximately 5000 ha have currently been cultivated [16] and progress is rather slow.

Although there is obviously a need for biomass and a potential for SRC, farmers hesitate to establish plantations on their agricultural land [21,22]. Several factors might explain this reluctance, e.g. a lack of expertise [23], a long-term commitment to a crop type with low flexibility to adapt to changing market conditions [24,25], uncertainties caused by political aspects like the discussed introduction of certification systems [26–28] or the risk caused by biological constraints such as plant diseases and pests [29]. However, possibly the most inhibiting factor might be the high investment costs combined with a delayed cash flow and unsure profitability in the future as in most cases, the decision to establish a SRC is driven by its economic prospects [16].

The aim of this study is therefore to analyse the economy of a typical SRC supply chain.

2. Material and methods

2.1. Analysed case

To obtain a complete picture, all relevant processes and materials are included into the economic analysis: soil

preparation, plant material and planting, weed control, harvesting and transport of the chips to the plant as well as the re-cultivation of the plantation.

As the most common solution for the harvesting operation, a forage harvester is assumed to be used which cuts and chips the trees in one working step. The fresh wood chips (50–60% moisture content, MC) are blown into an accompanying tractor-pulled trailer and are transported to an interim storage near the field (distance: 4 km). The capacity of the trailers is 20 cubic metre loose (m^3_{loose}). From the interim storage, chips are loaded by a wheel loader into a special truck (weight of 15 tons, maximum payload of 25 tons [30]) with trailers ($80 m^3_{\text{loose}}$) and transported to a heating plant at a distance of 50 km. The return is carried out empty as no back haulage is assumed.

This supply chain is defined as “basic chain”. Afterwards, four options are calculated to illustrate possible economic improvement of this basic SRC supply chain. All calculations refer to a total duration of the plantation of 20 years, which includes five rotations of four years each. A re-cultivation of the plantation to arable land is considered after 20 years.

2.2. Site location and data collection

In 2009, an experimental SRC of 4.5 ha was established with poplars (Max 4 and Monviso) in the mountainous region Schwäbische Alb (630 m above sea level) in southwest Germany close to the district of Sigmaringen (Baden-Württemberg) ($48^{\circ}6'N/09^{\circ}14'E$). The average soil quality index of the site is 37 [31], the average air temperature is $7.2^{\circ}C$ and precipitation is 790 mm per year on average (466 mm in the growing season). These conditions indicate a marginal growing situation. After soil preparation, the poplar cuttings were planted in a single row design. The distance between the rows is 250 cm and the distance between trees within a row is 60 cm, resulting in an initial planting density of 6700 trees per ha. A more detailed description of the plantation has been published by Aust (2012) [31].

Input data for this study were collected to a large extent on the experimental SRC plantation, e.g. the working time required for specific processes, costs of the cuttings, the amount and costs of herbicides used for soil preparation or the annual costs for land rent [32]. The required working time needs to be known in order to calculate the costs per hectare. It was measured during the operations (e.g. planting) for each process. This was done either with a stopwatch or with the amount of time billed by the contractor. Recorded scheduled times were in line with the values reported in the literature.

Detailed results on harvesting productivity were collected during numerous working time studies according to REFA (1991) [33]. The results have been published previously [34,35]. The average harvesting productivity reached was used to calculate the harvesting costs of the studied supply chain. Information about the working time needed to load the chips at the intermediate storage into the truck as well as for the transport were given by a local contractor [36] and information about the working time required for the re-cultivation of the SRC was taken from literature [37]. In a last step, yearly costs (e.g. land rent, Common Agricultural Policy (CAP)

subsidy payments) were considered, using original data from the experimental SRC.

2.3. Assessment of biomass yield

Within the project, continuous measurements and periodic field campaigns were conducted, to obtain the relevant tree parameters (tree dbh, height, dry matter and others) as inputs for the growth model MoBiLE-PSIM, which was used to assess the biomass yield for the overall stand duration of the plantation. The method applied is described in detail by Grote [38–40].

The average harvested volume of Max was used as a reference for the basic chain. It was calculated to be 30.5 Megagram of dry matter per hectare ($\text{Mg}_{\text{dm}} \text{ha}^{-1}$) on average in a four-year harvest cycle (Table 1). However, during the harvesting operations, biomass losses occur due to biomass remaining on the field. Taking these losses into account, the amount of marketable biomass at the gate of the heating plant was assumed to be 5% lower than the produced biomass (Table 1).

The amount of biomass corresponds to an average yield of $7.6 \text{ Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$. This result is rather low compared to yields reported in other European studies [41,42], but realistic, as the SRC was established on marginal agricultural land of a lower site class and with a low average temperature. Losses of biomass were also included. However, the sensitivity analysis performed considers a range of biomass yield figures per harvest corresponding to average yields between 7 and $14 \text{ Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$. These mostly higher yields were chosen to be analysed as it seems unrealistic to cultivate SRC if even lower yields are expected. Instead, the sensitivity analysis will show which biomass yield need to be reached to achieve financial results (annuities) that are attractive for the farmer (break-even point).

2.4. Market price of the wood chips

In 2012, the average market price for wood chips from SRC was reported to be $132.71 \text{ € Mg}_{\text{dm}}^{-1}$, based on 35% MC and excluding taxes [43]. However, a query of the purchase prices of local heating plants in south-western Germany resulted in a slightly lower price level ($120 \text{ € Mg}_{\text{dm}}^{-1}$) for wood chips from SRC with approx. 30% MC ($\triangleq \text{ca. } 12.9 \text{ GJ Mg}_{\text{dm}}^{-1}$) [32]. This relatively low MC can be achieved only if whole trees are cut and stored for drying for several weeks before chipping [44–46] (c.f. Section 3.2.4).

Table 1 – Biomass at the ProBioPa experimental site ($\text{Mg}_{\text{dm}} \text{ha}^{-1}$, per rotation).

Harvest no.	Mg_{dm} produced	Mg_{dm} delivered ^a
1	25.5	24.2
2	30.8	29.2
3	32.6	31.0
4	32.4	30.8
5	31.2	29.7
Average	30.5	29.0

^a 5% biomass is not recovered during harvesting which is considered as losses.

Today, in most cases SRC are harvested with a combined cut and chip system, using modified foragers equipped with special wood biomass headers [47]. In this case, the MC of fresh wood chips is up to 50–60% [47–50]. Accordingly, the heating value is lower (ca. $8.3 \text{ GJ Mg}_{\text{dm}}^{-1}$) which results in a lower market price ($90 \text{ € Mg}_{\text{dm}}^{-1}$).

The market price significantly influences the overall profitability. In addition to this, the chip market price is mostly locally defined and also depends on seasonal effects. Therefore, the performed sensitivity analysis considers alternative market prices for wood chips from SRC between 60 and $140 \text{ € Mg}_{\text{dm}}^{-1}$.

2.5. Calculation of production costs

The machine costs were calculated using the machine cost calculation scheme of the Food and Agriculture Organization of the United Nations (FAO) [51] on a full cost basis (excluding taxes). It includes the fixed (e.g. interest charges, depreciation, insurance, administration and lodging) and variable costs (e.g. fuel and lubricant, wages, repair and maintenance). Investments were assumed to be financed with outside capital (4% interest rate). The required data about the machines (e.g. purchase price, economic lifespan) were taken from the German Association for Technology and Structures in Agriculture [30] which provides one of the most comprehensive databases for equipment used in agricultural operations. The most relevant assumptions that needed to be made are shown in Table 2, the resulting machine costs in Table 3 and the respective costs per hectare in Table 4. Results also include costs for materials (e.g. herbicides, gasoline, etc.).

2.6. Calculation of annuities

A calculation model based on Excel (MS Office 2003) was developed to determine the production costs and the resulting annuities. As farmers usually consider the annual income if they evaluate whether the cultivation of SRC is favourable compared to common agricultural crops, the method of discounted cash flow (DCF) was applied. It integrates the effect of time on future inflows and outflows of cash by discounting to obtain their present value [42]. Therefore, the net present value (NPV) of the overall plantation was calculated and the annuity, which divides all costs and incomes into average annual values, was derived from the NPV. Respective formulas to calculate the NPV and the annuity were presented earlier in various studies [e.g. [42,16,32]].

Beside the income gained through selling the wood chips, yearly CAP subsidy payments (300 € ha^{-1}) were included in the calculations. An interest rate of 5% was considered for discounting. Furthermore, it was assumed that production costs will increase by 1.6% per year, which was the average inflation of the years 2000–2010 in Germany [52]. The market price for wood chips increased by 7–8% per year on average within the last years [31]. As it is difficult to anticipate if this development will continue in the next 20 years, a moderate increase in the market price of chips of 4% per year was assumed. All costs were calculated on a net basis (without taxes).

Table 2 – Input variables for machine cost calculations.

	Variable	Annual utilisation	Depreciation period ^a	Interest rate	Labour cost	Repair factor	Average fuel consumption
	Unit	h y ⁻¹	y	%	€ h ⁻¹	%	l h ⁻¹
Type of machine	Tractors (transport 1)	833	12	4	15	100	7.9
	Forage harvester	700	10	4	15	100	70
	Cutting head of forager	150	10	4	15	100	–
	Trailers	200	15	4	15	11	–
	Chipper	2000	10	4	15	90	32
	Truck (transport 2)	2000	6	4	15	90	34.5

^a Linear depreciation of machine purchase value.

3. Results

3.1. Distribution of costs and annuities

The total costs of a SRC in a 20 years cultivation is 17,564 € ha⁻¹. Thereof, harvesting and transport constitute the biggest share (55%, Fig. 1) followed by the land rental costs (21%) and costs for overhead and insurance (16%), while the costs for establishment (13%) and re-cultivation (5%) are rather low.

The resulting NPV is 863 € y⁻¹ ha⁻¹. Considering the current market price of 90 € Mg_{dm}⁻¹ this results in an annuity of 69 € y⁻¹ ha⁻¹ (Table 5).

The result is influenced by different parameters. As the share of land costs of the total costs was quite high (21%, Fig. 1) it was analyzed how this variable influences the annuity: If the rental costs were e.g. 150 € y⁻¹ ha⁻¹ (instead of 300 € y⁻¹ ha⁻¹) the annuity would increase to 219 € y⁻¹ ha⁻¹ (instead of 69 € y⁻¹ ha⁻¹). However, if higher land rents were charged, the annuity would be economical unfeasible (e.g. land rent costs of 500 € y⁻¹ ha⁻¹ lead to a negative annuity of minus 131 € y⁻¹ ha⁻¹).

Another parameter influencing the overall result is the height of the CAP subsidy. If it increased from 300 € y⁻¹ ha⁻¹ to

e.g. 500 € y⁻¹ ha⁻¹, the annuity would increase from 69 € y⁻¹ ha⁻¹ to 269 € y⁻¹ ha⁻¹ and thus would be competitive with annuities of annual market fruit cultivations on these site conditions [16]. If, on the other hand, the subsidy was abolished, the cultivation of SRC would lead to an annuity of –231 € y⁻¹ ha⁻¹.

Beside these two parameters a sensitivity analysis was carried out to analyse how the annuity varies, depending on the amount of harvested biomass per hectare and also on different market prices (Table 5).

The sensitivity analysis shows that, as expected, higher amounts of harvested biomass per hectare result in significantly higher annuities. It can be noticed that the increase of annuities is some kind of irregularly (Table 5) which can be explained by the transport process. Costs rise abruptly when a specific amount of biomass exceeds (80 m³_{loose}) and one more trip to the destination is necessary, although the trailer may not be fully loaded.

SRC with low biomass yields result in negative annuities and they are hardly economically profitable. However, higher biomass yields cannot be taken for granted on marginal land, even if these sites are regarded as first choice for SRC.

Another variable significantly influencing the economy of the SRC is the market price. Table 5 shows that it's impact is even higher than the biomass yield.

3.2. Options for improvements

If the rotation cycle were extended from 4 to 5 years, there would be more biomass output per hectare per harvest and fewer harvests would therefore be necessary when keeping the overall lifetime of the plantation to 20 years. If the SRC were established on dry land, site irrigation would perhaps improve the biomass yield.

On the other hand, higher market prices could be achieved if higher value were added to the product, e.g. by reducing the moisture content of the fresh wood chips via drying.

Therefore, four options (extended rotation cycle, irrigation, technical drying and natural drying in a two-step harvesting system), were analyzed. The respective findings are presented and discussed in the following sections.

3.2.1. Extension of the rotation cycle

If the rotation cycle were extended from four to five years, there would be more biomass per hectare per harvest. As a

Table 3 – Machine costs.^a

Type of machine	Costs	Unit
Tractor (67 kW)	36	€ h ⁻¹
Tractor (83 kW)	43	€ h ⁻¹
Forage harvester (pmh)	352	€ h ⁻¹
Forage harvester (non-pmh)	124	€ h ⁻¹
Whole rod harvester (pmh)	379	€ h ⁻¹
Whole rod harvester (non-pmh)	215	€ h ⁻¹
Truck	73	€ h ⁻¹
Trailer (20 m ³ _{loose})	0.55	€ t ⁻¹
Trailer (80 m ³ _{loose})	0.61	€ t ⁻¹
Wheeled loader	47	€ h ⁻¹
Chipper (pmh) ^b	119	€ h ⁻¹
Chipper (non-pmh) ^b	56	€ h ⁻¹

pmh = productive machine hour.

^a All costs without taxes.

^b A chipper is used in an alternative harvesting system presented in Section 3.2.4.

Table 4 – Cultivation costs (€ ha⁻¹).^a

Cost item	Costs	Frequency
Ploughing	104	n = 1
Grubbing	64	n = 1
Harrowing	68	n = 1
Weeding	264	n = 1
Planting	144	n = 1
Plant material	1675	n = 1
Re-cultivation	1800	n = 1
Weed control after harvest	77	n = 4
Harvesting ^b	555	n = 5
Transport 1 to interim storage ^b	188	n = 5
Transport 2 to plant (50 km) ^b	523	n = 5
Land rent	300	n = 20
Overhead	100	n = 20
Insurances	100	n = 20

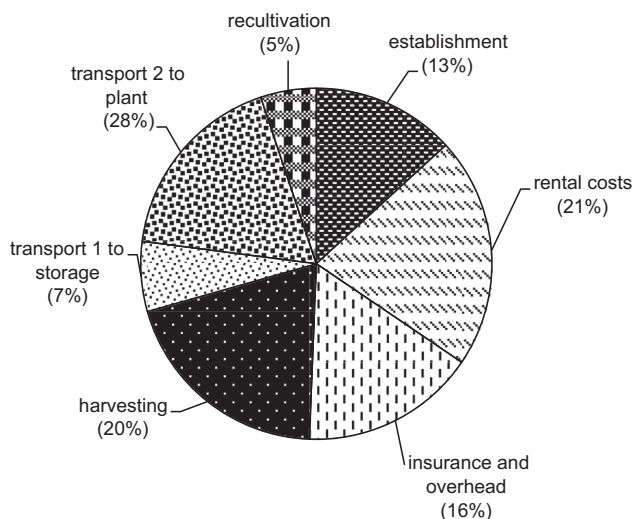
^a All costs without taxes.

^b Here, the average value of 5 rotations is reported only. In the calculations, the specific costs per harvests were considered.

consequence, the harvesting and transport costs per harvest would increase due to higher efforts (from 1266 € ha⁻¹ on average to 1430 € ha⁻¹ on average). At the same time, the amount of harvests within 20 years would be reduced. The resulting NPV would be 3182 € y⁻¹ ha⁻¹ (instead of 863 € y⁻¹ ha⁻¹) which would lead to an annuity of 255 € y⁻¹ ha⁻¹.

3.2.2. Irrigation

Within the framework of the project, a modern drip irrigation system (produced by Netafim Germany GmbH) was installed on the site. The specific investment costs for the irrigation system were extraordinary high, mainly due to the small size of the SRC (4.5 ha) and to experimental reasons. Usually, these kind of systems require an investment of about 2000 € ha⁻¹ and the annual costs for repair and maintenance over the lifetime of 20 years can be assumed to be 100 € y⁻¹ ha⁻¹ [32]. The latter (more realistic) investment costs were used for the calculation in this study. Annual costs were also incurred for the electrical power to run the irrigation system.

**Fig. 1 – Distribution of costs.**

Comparatively high amounts of power (1.5 kWh m⁻³) were needed to pump the water from a small river to the SRC site with a difference in height of 20 m on average.

An increase in biomass yield from 7.6 Mg_{dm} y⁻¹ ha⁻¹ to 10 Mg_{dm} y⁻¹ ha⁻¹ was assumed to account for the effect of irrigation. When calculating the respective annuities all other processes were kept identical compared to the basic chain (e.g. harvesting by forage harvester). The resulting annuity is minus 64 € y⁻¹ ha⁻¹ under the current market price of 90 € Mg_{dm}⁻¹. Only if an increase in SRC productivity from 7.6 Mg_{dm} y⁻¹ ha⁻¹ to 11 Mg_{dm} y⁻¹ ha⁻¹ could be achieved through irrigation would the annuity turn positive.

3.2.3. Drying of wood chips

The drying of fresh chips increases their heating value and thereby also the market price. However, fresh wood chips cannot just be dried in piles without risking certain problems that are well known and documented in the literature [50], e.g. self-heating and substantial biomass losses due to fungi and microbiological activities [53–55].

One possibility to dry the fresh chips efficiently is the use of surplus heat, e.g. from a biogas plant or other combustion processes. The required energy demand for the drying is 5.2 kWh m³⁻¹_{loose}. According to expert advice [56], costs of 10 € Mg_{dm}⁻¹ including fix and variable costs seem to be realistic for the drying of wood chips to a MC of 10–15% via the surplus heat of a biogas-based electricity generator. The process takes a drying time of a few days. A respective cost position was added to the periodic costs of the basic chain.

As a consequence of the drying a higher market price can be achieved for the wood chips (130 € Mg_{dm}⁻¹ because of 10–15% MC). The resulting NPV is 5004 € y⁻¹ ha⁻¹ and the resulting annuity is 402 € y⁻¹ ha⁻¹ (Table 6). The results show that the drying of chips can be economically very profitable if the added value can be turned into higher market prices (Table 6).

3.2.4. Harvesting of SRC with a whole rod harvester in a two-step operation

The results in Table 6 show that the drying of chips can be a favourable option. However, it is not always possible to use surplus heat from a plant or it might be unsuitable if large amounts of wood chips need to be handled. Therefore, another option to reduce the high MC of fresh chips is to employ a two-step harvesting system and thereby to use the rods' natural drying effect between the two steps of the operation.

As mentioned in Section 2.4, a relatively low MC (about 30%) can be achieved if whole trees are cut, collected and transported to the end of a row or to a defined place close to the field where the trees are stored for several weeks to dry before chipping. The Danish whole rod harvester "Stemster" is a machine operating according to this concept of two decoupled working steps. Time studies were carried out during harvesting operations with this system and its productivity was analysed [35]. One result was that, in contrast to usual forage harvesters, productivity does not seem to be significantly influenced by the amount of biomass per hectare [35]. Therefore, in this study identical productivities were assumed for all five harvests.

Table 5 – Annuities of cultivating SRC harvesting with a forage harvester.^a

Biomass per harvest ($\text{Mg}_{\text{dm}} \text{ha}^{-1}$) ($\hat{=}$ yield, $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	Market price ($\text{€ Mg}_{\text{dm}}^{-1}$)								
	60	70	80	90	100	110	120	130	140
26.6 ($\hat{=}$ 7 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–272	–177	–82	13	108	203	298	393	488
29.0 ($\hat{=}$ 7.6 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–240	–137	–34	69	172	276	379	482	585
30.4 ($\hat{=}$ 8 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–207	–98	11	119	228	336	445	554	662
34.2 ($\hat{=}$ 9 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–144	–22	100	222	344	467	589	711	833
38.0 ($\hat{=}$ 10 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–113	23	159	294	430	566	702	838	973
41.8 ($\hat{=}$ 11 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–45	104	254	403	552	702	851	1000	1150
45.6 ($\hat{=}$ 12 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	19	182	345	508	671	834	997	1160	1323
49.4 ($\hat{=}$ 13 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	52	228	405	582	758	935	1111	1288	1464
53.2 ($\hat{=}$ 14 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	121	311	501	691	881	1071	1261	1451	1645

^a Total duration: 20 years (5 harvests).

Respective harvesting as well as the chipping costs were calculated (harvesting costs 645 € ha^{-1} and chipping costs 435 € ha^{-1} on average per harvest). All other costs were calculated as comparable to the basic chain (one-step harvesting operation with forage harvester, Table 4). To account for the higher logistical complexity in the decoupled two step system, a delay factor of 30% was considered for the total working time of the chipping operation [57]. The storage on the field was assumed to be free of charge. Furthermore, it was not considered that capital in the form of biomass has being tied in the field for approx. four months.

With the two-step harvesting system wood chips with ca. 30% MC are produced which leads to a market price of about $120 \text{ € Mg}_{\text{dm}}^{-1}$. The resulting NPV is $3593 \text{ € y}^{-1} \text{ha}^{-1}$ and the respective annuity is $288 \text{ € y}^{-1} \text{ha}^{-1}$ (Table 7).

This result is lower compared to the option for technical drying of the chips with surplus heat after a one-step harvesting operation with a forage harvester (Table 6), but might be an alternative in situations where technical drying after the one-step harvesting operation is not feasible (Table 5).

4. Discussion and conclusion

For economic evaluation and comparison, a basic SRC supply chain was analysed where a state-of-the-art harvesting technology was applied: A standard forage harvester with a

special wood biomass header producing fresh wood chips (50–60% MC) being delivered via truck to a plant at a distance of 50 km. In order to avoid competition with land that is well suited for food production, only marginal agricultural land was assumed to be used for the cultivation of SRC. This land is a very important land resource for bioenergy production, but the conditions are economically not favourable.

The result shows that with biomass yields below $7\text{--}8 \text{ Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$, which are typical for marginal land, this widely used supply chain is hardly profitable (Table 5). This result is in line with the findings of earlier studies [42,16] and might be an explanation for the hesitant establishment of SRCs in Germany.

It has to be stressed that the most critical point of this study is the reliability of the data, as different sources have been used. This was done in order to get as realistic data as possible. E.g., the cost for land rent might seem to be rather high, but it is an average value representative for Germany [58] and it is also comparable to the findings of other studies [42]. However, the sensitivity of the results was tested by changing the values of crucial parameters (e.g. the height of CAP subsidy payments).

Taking the CAP subsidy payments by the EU as an economic benchmark (around 300 € ha^{-1}), farmers will hardly change to perennial SRC when expected annuities are that low. The average annuities of annual market fruit cultivations are between $226 \text{ € y}^{-1} \text{ha}^{-1}$ on lower and $462 \text{ € y}^{-1} \text{ha}^{-1}$ on

Table 6 – Annuities of cultivating SRC when including technical drying of wood chips.^a

Biomass per harvest ($\text{Mg}_{\text{dm}} \text{ha}^{-1}$) ($\hat{=}$ yield, $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	Market price ($\text{€ Mg}_{\text{dm}}^{-1}$)								
	60	70	80	90	100	110	120	130	140
26.6 ($\hat{=}$ 7 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–346	–251	–156	–61	34	129	224	319	414
29.0 ($\hat{=}$ 7.6 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–321	–218	–114	–11	92	195	298	402	505
30.4 ($\hat{=}$ 8 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–291	–183	–74	35	143	252	361	469	578
34.2 ($\hat{=}$ 9 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–231	–117	5	127	249	372	494	616	738
38.0 ($\hat{=}$ 10 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–222	–87	49	185	321	456	592	728	864
41.8 ($\hat{=}$ 11 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–165	–16	134	283	432	582	731	880	1030
45.6 ($\hat{=}$ 12 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–108	55	218	381	544	707	870	1033	1196
49.4 ($\hat{=}$ 13 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–89	87	264	440	617	793	970	1146	1323
53.2 ($\hat{=}$ 14 $\text{Mg}_{\text{dm}} \text{y}^{-1} \text{ha}^{-1}$)	–31	159	349	539	729	919	1110	1300	1490

^a Total duration: 20 years (5 harvests).

Table 7 – Annuities of SRC harvesting with a whole rod harvester.^a

Biomass per harvest (Mg _{dm} ha ⁻¹) (≡ yield, Mg _{dm} y ⁻¹ ha ⁻¹)	Market price (€ Mg _{dm} ⁻¹)								
	60	70	80	90	100	110	120	130	140
26.6 (≡ 7 Mg _{dm} y ⁻¹ ha ⁻¹)	-352	-257	-162	-67	28	123	218	314	409
29.0 (≡ 7.6 Mg _{dm} y ⁻¹ ha ⁻¹)	-331	-228	-124	-21	82	185	288	392	495
30.4 (≡ 8 Mg _{dm} y ⁻¹ ha ⁻¹)	-289	-180	-72	37	146	254	363	471	580
34.2 (≡ 9 Mg _{dm} y ⁻¹ ha ⁻¹)	-226	-104	18	140	263	385	507	629	752
38.0 (≡ 10 Mg _{dm} y ⁻¹ ha ⁻¹)	-163	-27	108	244	380	516	651	787	923
41.8 (≡ 11 Mg _{dm} y ⁻¹ ha ⁻¹)	-136	13	163	312	461	611	760	910	1059
45.6 (≡ 12 Mg _{dm} y ⁻¹ ha ⁻¹)	-73	90	253	416	579	742	905	1067	1230
49.4 (≡ 13 Mg _{dm} y ⁻¹ ha ⁻¹)	-10	166	343	519	696	872	1049	1225	1402
53.2 (≡ 14 Mg _{dm} y ⁻¹ ha ⁻¹)	17	207	397	587	777	968	1158	1348	1538

^a Total duration: 20 years (5 harvests).

medium site conditions [16]. However, as the farmer is familiar with the annual cultivation system there is no incentive to change the current system towards SRC from his point of view.

Results show that harvesting and transport constitute the biggest share of the overall cultivation costs (Fig. 1). The costs of the transport via truck and trailer mainly depend on the transport distance (50 km one way). As a consequence, chips should preferably be used locally. The harvesting costs depend significantly on the productivity of the harvesting machine which increases with increasing amounts of biomass per hectare until technical restrictions due to limitation in diameter are reached [34].

An option to improve the revenues might be to extend the rotation cycle from 4 to 5 years (c.f. Section 3.2.1). This would also reduce cultivation costs due to fewer harvests (if the overall lifetime of the plantation of 20 years is kept) and thereby lead to higher annuities (255 € y⁻¹ ha⁻¹ instead of 69 € y⁻¹ ha⁻¹). Technical restrictions (e.g. a tree diameter which exceeds the capacity of the feeder head of the forage harvester) may put a limit to the extension of the rotation period. Currently, machine development is ongoing and the trend is towards upsized forager-based harvesters [59].

In general, more attractive annuities can be expected if biomass yields can be improved. Results show that the investment in a modern drip irrigation system in order to increase the yield is not profitable. Probably, a more common and less expensive irrigation system (e.g. centre pivot) and water supply by gravity without electric pumps would be key to producing more favourable economic results. These alternatives might increase in importance as many studies show that the plant-available water balance is the most important site factor influencing the SRC incremental growth rates [60–64] and that—especially poplars—might develop well on marginal land as long as there is no limitation in water supply.

Not included in this study is a possible increase of the biomass yield through poplar plant breeding programs currently ongoing in Europe and North America.

What turned out to be a key factor is the market price that can be achieved for the wood chips. Results show that it has an even higher impact on annuities than the yield. In this context, a very promising option is the technical drying of wood chips in order to increase the market price (Table 6). Farmers should not only aim to harvest at low cost and deliver

fresh chips from their SRC, but also try to optimise the whole supply chain by including drying as a value adding process. Currently, many studies are under way which analyse the effectiveness and the costs of different drying and storing concepts and techniques, e.g. the “dome aeration technology” [65]. Using an alternative two-step harvesting system (Table 7) that allows the natural drying of the rods can also be economically attractive, even if the harvesting process as such is less productive (11 instead of 21 Mg_{dm} pmh⁻¹). If the technical drying of chips is not feasible for the farmer, a system like the analysed cut and storage system is a good alternative (if the market pays higher prices for dry chips).

It can be concluded that the average results for the cultivation of SRC on marginal land is lower than the CAP subsidy payments granted to farmers by the EU. Calculations showed that if these payments increased from 300 € y⁻¹ ha⁻¹ to 500 € y⁻¹ ha⁻¹, the annuity would increase from 69 € y⁻¹ ha⁻¹ to 269 € y⁻¹ ha⁻¹ (c.f. Section 3.1) and thus be competitive with annuities of annual market fruit cultivations under these site conditions. As a consequence, if it is a political objective to significantly extend the SRC plantations on marginal land, the government should increase the subsidy payments. This is true until there will be reliable options to improve biomass yields or unless the chip market price increases to a level which allows the profitable cultivation of SRC. Furthermore, there should be a focus on decentralised energy supply systems with short transport distances (which reduces transport costs and thereby increases revenues). As a positive side effect, local energy systems would constitute new employment and income sources for rural areas [66]. In a broader focus, replacing fossil energy sources by biomass from SRC also leads to lower environmental impacts as LCA studies show [67,68,32]. Avoided emissions are higher compared to annual crops. This should be taken into account by the government when supporting climate policy through the instrument of subsidies.

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